

Intermittency in superfluid helium counterflow turbulence

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(Dated: April 21, 2026)*

INTRODUCTION

The chaotic dynamics of turbulent fluids is a ubiquitous phenomenon dominating a vast array of physical systems in nature [1]. Turbulence shapes the patterns and properties of fluids of all kinds, from ordinary viscous fluids in Navier-Stokes turbulence [2] to electrically conducting fluids in magneto-hydrodynamics [3] and quantum fluids in quantum turbulence [4, 5]. Such systems are characterised by the existence of a wide range of length scales across which inviscid conserved quantities, such as energy, are transferred without any loss in the spirit of the cascade described by Richardson [6]. In 3D classical turbulence this process is a direct cascade where energy is transferred from large to small scale structures at a constant rate of energy with a well-known scaling law of Kolmogorov's phenomenological K41 theory [7]. The dynamics of these filament structures can result in collisions known as vortex reconnections, altering the topology of the system and re-distributing energy between scales and fluid components [8, 9]. The richness of turbulence in quantum fluids arises from the additional length scale ℓ , separating the dynamics at large scales where the flow is unaffected by small-scale vortex structures, and small scales where the flow is strongly dependent on them. At these large scales, the same cascade phenomenology of the K41 theory has been observed in quantum fluids [10], and the turbulent flows have been known to behave intermittently [5]. Velocity increments $\delta\mathbf{v}(\mathbf{r}) = \mathbf{v}(\mathbf{x}+\mathbf{r}) - \mathbf{v}(\mathbf{x})$ and corresponding structure functions $S_p = \langle |\delta\mathbf{v}(\mathbf{r})|^p \rangle$ recover scaling exponents $S_p \sim r^{\zeta_p}$ in the inertial range with striking similarity to classical turbulence [11, 12].

Recent work in classical turbulence [13] and quantum turbulence [14–17] have demonstrated the success of velocity circulation Γ given by

$$\Gamma_r(\mathbf{v}) = \oint_{\mathcal{C}(r)} \mathbf{v} \cdot d\mathbf{r} = \iint_{\mathcal{A}(r)} (\nabla \times \mathbf{v}) \cdot \hat{\mathbf{n}} dA, \quad (1)$$

where $\mathcal{C}(r)$ denotes a closed loop enclosing an around area $\mathcal{A}(r)$, as an alternative to velocity increments for the analysis of intermittency. For higher order moments,

high-Reynolds number simulations have demonstrated that circulation scaling exponents are well described by monofractal fits [13], exhibiting a simpler statistical structure in the inertial range than the multi-fractal scaling of velocity increments. In fact, Gross-Pitaevskii (GP) simulations [14, 15] and finite-temperature grid turbulence in ^4He [16] display a scaling of circulation variance $\langle |\Gamma_r|^2 \rangle \sim r^{8/3}$ within the inertial range, coinciding with classical turbulence [1].

In this Letter, we compute the velocity circulation statistics for steady-state counterflow turbulence in ^4He and compare our findings with a purely superfluid Taylor-Green simulation. Counterflow turbulence differs significantly from mechanically generated turbulence in two key aspects. Firstly, the imposed counterflow velocity facilitates an energy injection mechanism by the Donnelly-Glaberson instability [8, 18]. Secondly, steady-state counterflow turbulence is known to be strongly anisotropic, and has a tendency to develop large-scale quasi-two-dimensional structures [15, 19] which can support an inverse cascade of energy [20]. To account for this anisotropy, we split our analysis of the counterflow state into the orientations parallel and perpendicular to the counterflow direction.

METHOD

Superfluid dynamics are modelled by employing the recently developed FOUCAULT model [21]. We follow Schwarz's approach [22], by exploiting the large separation of scales between the ^4He vortex core size a_0 and the average inter-vortex spacing ℓ . We parametrize the superfluid vortex lines as 1D space curves $\mathbf{s}(\xi, t)$ with ξ and t the arclength and time respectively. The corresponding equation of motion is

$$\dot{\mathbf{s}}(\xi, t) = \mathbf{v}_{s\perp} + \beta \mathbf{s}' \times \mathbf{v}_{ns} + \beta' \mathbf{s}' \times [\mathbf{s}' \times \mathbf{v}_{ns}] \quad (2)$$

where $\mathbf{s}' = \partial\mathbf{s}/\partial\xi$ is the unit tangent vector at $\mathbf{s}$, $\mathbf{v}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ where $\mathbf{v}_n$ and $\mathbf{v}_s$ are the normal fluid and superfluid velocities at $\mathbf{s}$, and β, β' are temperature

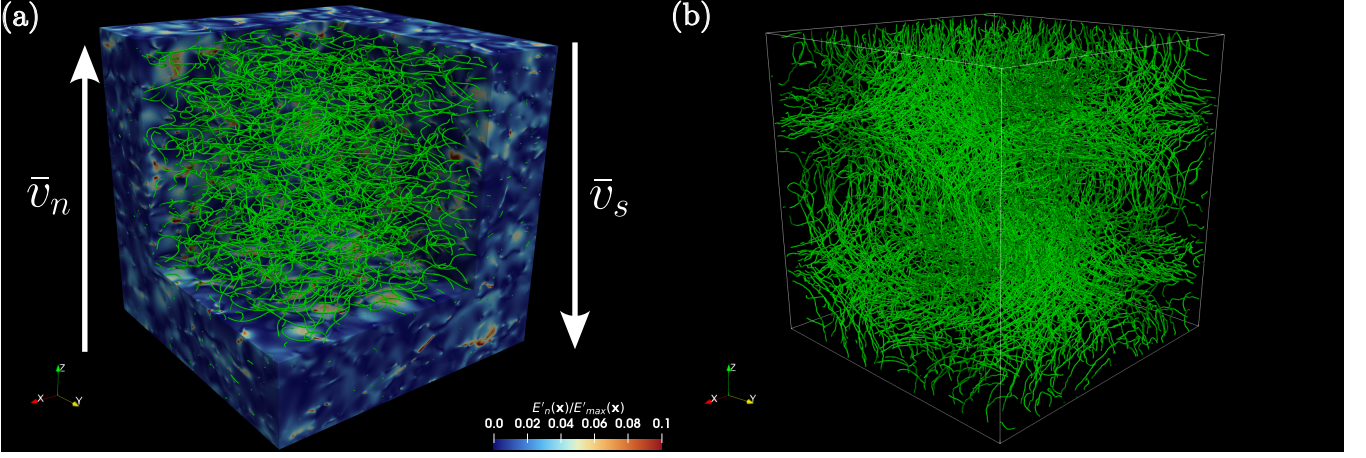


FIG. 1: Visualisation of turbulent vortex tangles. (a) Counterflow-induced turbulence generated at $T = 2.1$ K with a counterflow velocity of $v_{ns} = \bar{v}_n - \bar{v}_s = 0.94$ cm/s. The blue and red surface rendering shows the normalised kinetic energy density of normal fluid fluctuations E'_n/E'_{max} where $E'_n = |\mathbf{v}_n - \bar{\mathbf{v}}_n|^2$. (b) Turbulence generated by an initial Taylor-Green configuration (see Supplementary Materials for details) $T = 0$ K. In both (a) and (b) superfluid vortex lines are represented as green tubes with exaggerated size.

and Reynolds number dependent mutual friction coefficients. The superfluid velocity $\mathbf{v}_s$ is computed via a desingularized Biot-Savart integral, accelerated using the tree algorithm [23] (see Supplementary Materials). We describe the normal fluid in a classical way, using the incompressible ($\nabla \cdot \mathbf{v}_n = 0$) Navier-Stokes equation

$$\frac{\partial \mathbf{v}_n}{\partial t} + (\mathbf{v}_n \cdot \nabla) \mathbf{v}_n = -\frac{1}{\rho} \nabla p + \nu_n \nabla^2 \mathbf{v}_n + \frac{\mathbf{F}_{ns}}{\rho_n} \quad (3)$$

where ρ_n and ρ_s are the normal fluid and superfluid densities, $\rho = \rho_n + \rho_s$, p is the pressure, ν_n is the kinematic viscosity of the normal fluid and $\mathbf{F}_{ns}$ is the mutual friction force per unit volume.

In this study, we prepare two distinct turbulent vortex tangles generated by (a) a $T = 2.1$ K thermal counterflow (CF) with an average inter-vortex distance ℓ_{CF} and (b) a $T = 0$ K evolution of a tangle with a Taylor-Green initial condition (TG) with an average inter-vortex spacing ℓ_{TG} . Both of these tangles are visualized in the left and right panels of Fig. 1 respectively. The velocity circulation statistics of $\mathbf{v}_s$ are obtained by a vorticity coarse-graining process described in the Supplementary materials. Furthermore, the statistics related to the counterflow are decomposed into the normal fluid (NF) and superfluid (SF) components. Each of these components are further decomposed into the perpendicular ($\perp$) and ($\parallel$) components, corresponding to the directions over which statistics are averaged. Perpendicular (parallel) relates to the statistics averaged in the direction where the flux of vorticity through a square loop of size r is perpendicular (parallel) to the counterflow velocity $\mathbf{v}_{ns}$.

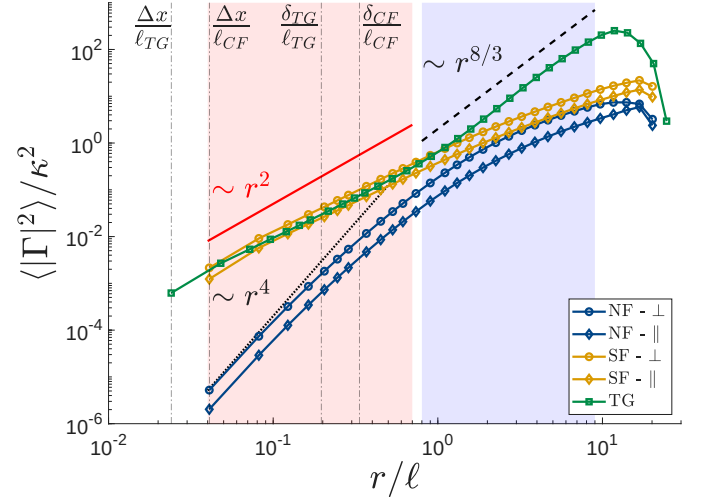


FIG. 2: Circulation variance $\langle |\Gamma|^2 \rangle$ around square loops of size r . The blue and yellow curves show the normal fluid and super fluid simulations respectively, which are subdivided into the perpendicular (circles) and parallel (diamonds) components. The green squares show the Taylor-Green simulation.

RESULTS

Firstly, we begin by considering the simplest observable for velocity circulation statistic - the variance $\langle |\Gamma|^2 \rangle$ for different loop sizes r . The circulation variance scalings are shown in Fig. 2, where we have identified two scaling regimes, a quantum regime $r \ll \ell$ with red shading and a classical regime $r \gtrsim \ell$ with blue shading. Indeed in the quantum regime we find good agreement for the superfluid velocity statistics with the small-scale theory

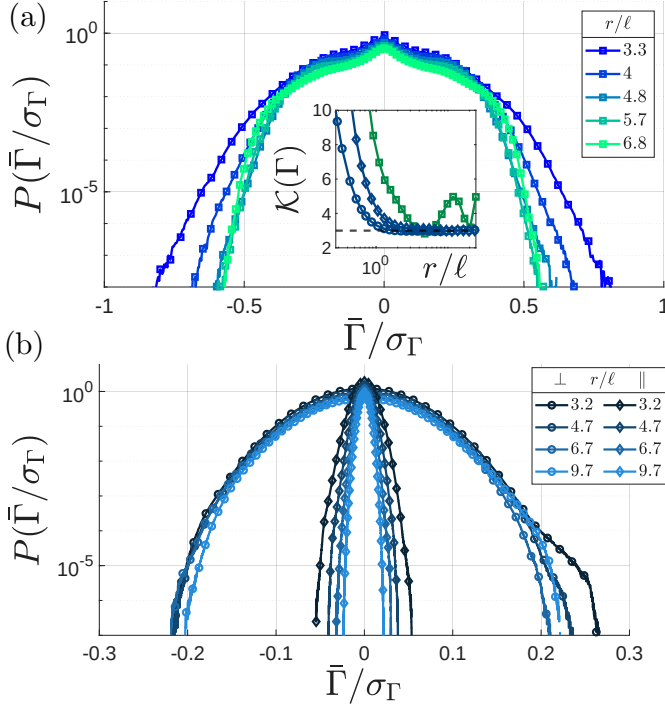


FIG. 3: PDFs of the velocity circulation for different loop sizes $r/\ell > 1$, scaled by the standard deviation σ_Γ . (a) shows the Taylor-Green simulation, while (b) shows both of the normal fluid parallel (diamonds) and perpendicular (circles) components. The inset of (a) shows the flatness $\mathcal{K}(\Gamma) = \langle |\Gamma|^4 \rangle / \sigma_\Gamma^4$, with the same color scheme as in Fig. 2. The black dashed line marks the Gaussian flatness of $\mathcal{K} = 3$. See the Supplementary materials for the SF PDFs.

$\langle |\Gamma|^2 \rangle \sim r^2$ as discussed in Ref. [14] using the Gross-Pitaevskii (GP) model. We observe that in this regime the scaling not only follows the GP theory, but the quantum region scaling is independent of thermal effects and the nature of the turbulent driving force. Similarly, the normal fluid also exhibits the well known viscous scaling of $\langle |\Gamma|^2 \rangle \sim r^4$ for small loop sizes r , as a result of smoothness at very small scales [13].

The cross-over from the quantum $r \ll \ell$ to the classical $r \gtrsim \ell$ marks a stark contrast in the superfluid circulation variance scaling. While the TG scales in accordance with a partial polarization of vortices where $r^{8/3}$ [24], the CF scaling, appears to follow a scaling of $\sim r^{1.58}$, a weaker scaling than for a fully random tangle of vortex lines which exhibits r^2 scaling in the classical range. Remarkably both components of the SF and NF statistics of CF appear to adhere to this scaling law for loops $r \gtrsim \ell$. For the remainder of this paper, we will only focus on the classical range where the loop sizes $r > \ell$.

In fully-developed isotropic classical turbulence, velocity circulation statistics adhere to a strictly non-Gaussian distribution with long tails [25]. Deviations from Gaus-

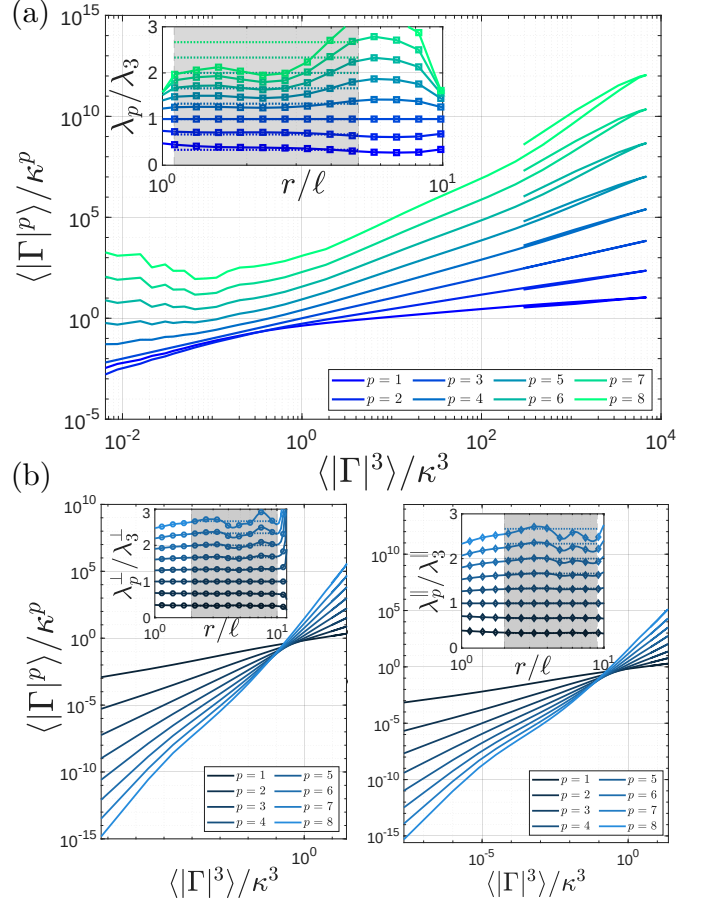


FIG. 4: Circulation moments $\langle |\Gamma|^p \rangle$ in extended self-similarity (ESS) using $p = 3$. (a) shows the Taylor-Green simulation, while the left and right panels of (b) are the normal fluid perpendicular and parallel component respectively. The insets of each plot give the local slope, where $\lambda_p/\lambda_3 = d[\log \langle |\Gamma|^p \rangle] / d[\log \langle |\Gamma|^3 \rangle]$ and the horizontal lines correspond to the K41 scalings $\lambda_p^{K41} = 4p/3$. The shaded regions indicate the classical region where the data was used to estimate the exponents.

sian statistics are the hallmark of intermittency, and represent a deviation from self-similar scaling which are evident in the TG simulations shown in Fig. 3a. The circulation PDF of the NF components differs very strongly (see Fig. 3b). Strikingly, we see a complete collapse to Gaussian statistics for the perpendicular components, while the parallel component exhibits very small deviations. We show this collapse by computing the flatness $\mathcal{K}(\Gamma) = \langle |\Gamma|^4 \rangle / \sigma_\Gamma^4$, where σ_Γ is the standard deviation $\sqrt{\langle |\Gamma|^2 \rangle}$. The flatness is displayed in the inset of Fig. 3a. In the classical regime the NF components monotonically decay to the Gaussian value of $\mathcal{K} = 3$, while the non-Gaussian TG simulation deviates in the same region. Indeed we observe that the presence of quantized vortices suppresses the intermittent behaviour of the classically-

driven NF.

As is customary in the classical turbulence, we will characterize the intermittency by measuring the deviation of the circulation moments of order p given by $\langle |\Gamma|^p \rangle$ from the self-similar K41 scaling. The TG and NF moments are shown in Fig. 4, where we have plotted the circulation moments in extended self-similarity (ESS) using the third-order circulation moment $\langle |\Gamma|^3 \rangle$. In this way, we define the scaling exponents λ_p as

$$\langle |\Gamma|^p \rangle \sim [\langle |\Gamma|^3 \rangle]^{(\lambda_p/\lambda_3)p} \quad (4)$$

In order to compute the scaling exponents, we find the local slope of the moments in ESS, which is given by

$$\left(\frac{\lambda_p}{\lambda_3} \right) (r) = \frac{d[\log \langle |\Gamma|^p \rangle]}{d[\log \langle |\Gamma|^3 \rangle]} \quad (5)$$

As the local-slope of pure power-law functions is flat, we look for flat regions where the loop sizes are $r > \ell$, and then average over the selected regions. The scaling exponents λ_p are shown in Fig. 5 for integer moments of order p . As expected, the TG scaling exponents lie closely to the bifractal behaviour identified in high Reynolds number classical turbulence [13] and also in pure GPE quantum turbulence [14]. The bifractal fit for classical turbulence intermittency uses a robust linear scaling $\lambda_p = \alpha p$ where $\alpha \approx 1.367$ for low order moments $p < 3$, which is in good agreement with our findings. For larger moments, we see that both the TG and VF circulation moments can be closely related to the monofractal fit $\lambda_p = hp + (3 - D)$ with fractal dimension D and Hölder exponent h with weak Reynolds number dependence. For the bifractal fit used in Fig. 5, we used $D \approx 2.2$ and $h \approx 1.1$, which correspond to the values estimated in the highest Reynolds number flow in Ref. [13].

We make two major remarks about CF scaling exponents. Firstly, we can see a striking departure between the NF and VF statistics. While the VF components closely follow the intermittent monofractal scaling for larger moments, the NF components exhibit a collapse to self-similar K41 theory. Secondly, in both the NF and VF statistics, the parallel component shows a higher-level of intermittency than the perpendicular component. Indeed in the inset of Fig. 5, we can see from the proportional deviation from K41 theory that the NF perpendicular component almost perfectly converges to self-similarity, while the parallel component exhibits a small level of intermittency. Within ESS, a linear monofractal of the form above is under-determined, and therefore we can only estimate the ratio $a = h/(3 - D)$, which for the NF parallel component gives $a \approx 9.34$. In order to describe the velocity circulation intermittency which we observe in the NF parallel component we adopt the She-Leveque log-Poisson model [26], such that the scaling exponents are given by $\lambda_p = \alpha p + C [1 - \beta^{2/3}]$ where $C = 3 - D$

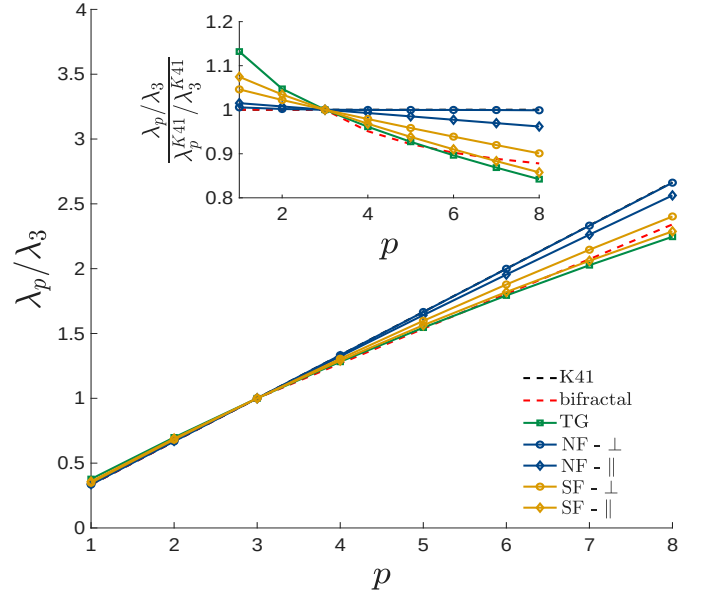


FIG. 5: Scaling exponents of velocity circulation moments, estimated within the classical range $r > \ell$. The colour scheme and line styles is the same as in Fig. 2. The black dashed line represents the self-similar K41 scaling $\lambda_p^{K41} = 4p/3$. The inset shows the proportional deviation from K41 theory.

is the codimension of the most singular structures and β characterizes the intermittency.

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